

Amrit Mahajan

(480) 527-3622 | amrit.mahajan007@gmail.com | [linkedin.com/in/amrit-mahajan](https://www.linkedin.com/in/amrit-mahajan) | amrit-mahajan.vercel.app | Open to Relocation

SUMMARY

Data Scientist with 2.5+ years of experience designing ML & Generative AI solutions for financial platforms serving 40M+ users. Delivered measurable business impact through a churn model (+3.8% retention), a notification engine (+25% CTR), and a hybrid RAG pipeline (-85% latency). Skilled in owning the ML lifecycle end-to-end, from data exploration to production deployment.

EDUCATION

Arizona State University, Tempe, AZ

Aug. 2024 - May 2026

Master of Science, Computer Science

GPA: 4.0/4.0

Coursework: Data Mining, Statistical ML, Natural Language Processing, Web Mining, Cloud Computing, Data Processing at Scale

Vellore Institute of Technology, India

Jul. 2018 - May 2022

Bachelor of Technology, Computer Science Engineering | Merit Scholarship Recipient (Top 10 rank)

GPA: 9.29/10

Coursework: Data Structures and Algorithms, Operating Systems, Database Management, Software Design, Computer Architecture

EXPERIENCE

Wells Fargo | Gen AI Intern, Quantitative Analytics Program

Jun. 2025 - Aug. 2025

- Built a Generative AI question-answering system for bankers using enterprise customer data, enabling context-aware insights and reducing client preparation time by 30% in pilot testing.
- Designed and tested a hybrid Retrieval Augmented Generation (RAG) pipeline using BM25 and FAISS to replace long-context prompting, reducing retrieval latency by 85% through targeted chunk retrieval and smaller-model inference.
- Developed an LLM evaluation framework using DeepEval and DSPy to benchmark and optimize prompt engineering strategies, incorporating human-in-the-loop evaluation, LLM-as-a-judge, and adversarial testing to assess model reliability and safety.

Bajaj Finance Limited | Data Scientist

Jul. 2022 - Jul. 2024

- Led development and deployment of a Churn Prediction model on the Bajaj Finserv App (40M+ users), using SHAP-based explainability to drive retention strategies that increased customer retention by 3.8%.
- Built a Notification Recommendation Engine combining time-series clickstream data and a business rules engine, increasing CTR by 25% and eliminating manual campaign execution effort across 40 business teams.
- Scaled experimentation from 20 to 120+ A/B tests per quarter by designing a self-serve testing framework in Python, cutting analyst review time by 65% and onboarding 25+ stakeholders.
- Pioneered MLOps adoption by implementing MLflow in Databricks, reducing model deployment effort by 50%, standardizing experiment tracking, and driving adoption across 3 teams.
- Reduced forecast error (MAPE) by 20% by developing LSTM models with attention mechanisms for long-range forecasting.

Bajaj Finance Limited | Data Science Intern

Jan. 2022 - Jul. 2022

- Built automated forecasting pipelines in Pandas and PySpark for app metrics (MAU, DAU, traffic, and net users), reducing monthly reporting time from 5-6 days in Excel to 1 day.
- Segmented users by notification behavior using tree-based models, then partnered with the product team to translate findings into governance rules that reduced notification-driven churn risk.

PROJECTS

Identifying LLM Performance Gaps Across Languages | Research Project at ASU

- Evaluated multilingual performance gaps in LLMs using OLMo models across 100+ languages and 5 reasoning domains (Math, Reasoning, NLI, Common sense, healthcare) using high-quality translation pipelines.
- Quantified a 0.8 rank correlation between language-token exposure and cross-lingual accuracy, isolating it as the strongest structural driver among parallel data and script effects, to guide fairer LLM development.

Fraud Detection System | Outstanding Project Award, CSE 598 Data-Intensive Computing for ML, ASU

- Built an end-to-end fraud detection pipeline by integrating a CNN into Meta's Velox engine with custom C++ functions, enabling real-time SQL-based inference and achieving 100% accuracy on 1,000 IDNet images.
- Deployed on AWS EC2 with Docker, benchmarking Velox vs. PyTorch (CPU & GPU) to show 4x faster query execution and supporting fairness analysis across gender, age, and ethnicity.

Grocery Agent | LangGraph, Neo4j, Playwright, LLM Tool-Calling (Python)

- Engineered an autonomous grocery agent with LangGraph orchestrating a ReAct tool-calling loop that searches products, validates diet safety, and swaps in substitutes to build checkout-ready carts.
- Built a Neo4j knowledge graph (1.16M+ nodes, 12.5M+ relationships) from Open Food Facts for deterministic diet validation, then automated cart checkout with Playwright, validated by 37 passing tests.

TECHNICAL SKILLS

Programming & Data: Python, SQL, PySpark, Numpy, Pandas, C++, Excel, PowerBI

Modeling & ML: Statistics, Predictive Modeling, Time Series, Deep Learning, NLP, Scikit-learn, TensorFlow

Generative AI: LLMs, Prompt Engineering, RAG, FAISS, LangChain, LangGraph, Hugging Face, Agentic Workflows, Claude Code

MLOPs & Cloud: Databricks, MLflow, Docker, AWS, Git

Certifications: Generative AI with LLMs, ML Engineering for Production (MLOps) Specialization, TensorFlow Developer Professional Certificate, Deep Learning Specialization (DeepLearning.AI), AWS Certified Cloud Practitioner